MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY, RELAY SENSE AND K12 OPERATION

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-60898 comprise the complete requirements for the Relay Sense and K12 Operation Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-60898	Circuit Card Assemblies, General Specification for

DRAWINGS

U. S. Army, Frankford Arsenal

F10554179	Electronic Component Assembly, Relay Sense and K12 Operation

PACKAGING DATA SHEET

10554179	Packaging of Circuit Card Assembly, Relay Sense and K12 Operation

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:04Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Fabrication. The assembly shall be manufactured in accordance with Drawing F10554179 and drawings pertaining thereto.

3.2 General specification. The assembly shall meet the requirements of MIL-C-60898.

3.3 Environmental service conditions. See MIL-C-60898.

3.3.1 Performance at temperatures extremes. For the conditions of 3.3.2 of MIL-C-60898, the requirements of 3.4 shall be met with exceptions as specified in the following subparagraphs.

3.3.1.1 Logic “1” output. An output logic “l” shall be greater than +2.5 vdc but less than +4.4 vdc.

3.3.1.2 Logic “O” output. An output logic “O” shall be greater than +0.01 vdc but less than +0.4 vdc.

3.3.1.3 Pin 2 output. The high and low output voltage of pin 2 shall be +18.0 ± 1.0 vdc and greater than +0.01 vdc but less than +1.2 vdc, respectively.

3.3.1.4 “High” output. A "high" output shall be +20.5 ± 1.5 vdc.

3.3.1.5 “Low” output. A “low” output shall be greater than +0.01 vdc but less than +1.2 vdc.

3.4 Electrical performance

3.4.1 Conditions of operation. The requirements of 3.4.2 and 3.4.3 shall be met with the termination to the board as shown in Figure 1. A logic “O” input shall be +0.50 ± 0.01 vdc and a logic “l” input shall be +2.80 ± 0.01 vdc.

3.4.2 Logic outputs. An output logic “l” shall be greater than +2.8 vdc but less than +4.4 vdc. An output logic “O” shall be greater than +0.01 vdc but less than +0.4 vdc. A high output shall be +21.5 ± 0.5 vdc and a low output shall be greater than +0.01 vdc but less than +1,2 vdc. Pin 1 is ground.

3.4.2.1 Output pin 27. With the inputs applied respectively as specified in Table I, the output at pin 27 shall be either “high” or “low” as specified in Table 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:04Z
Check the source to verify that this is the current version before use.

Table I

Input Pins		Output Pin
22	23	27
0	0	high
0	1	high
1	0	low
1	1	high

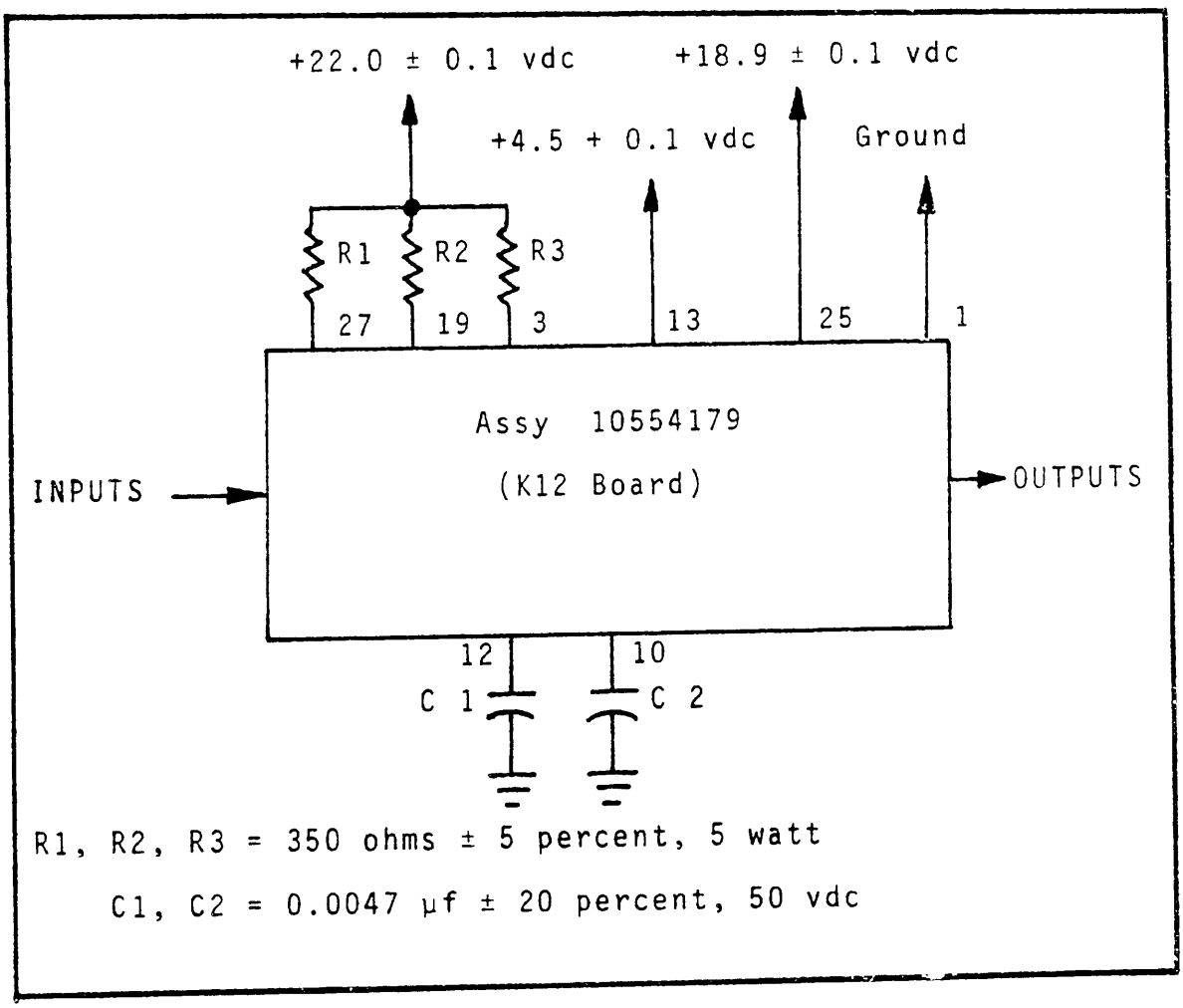


Figure 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:04Z
Check the source to verify that this is the current version before use.

3.4.2.2 Output pin 19. With the inputs applied respectively as specified in Table 11, the output at pin 19 shall be either “high” or “low” as specified in Table II.

Table II

Input Pins		Output Pin
20	21	19
0	0	high
0	1	high
1	0	low
1	1	high

3.4.2.3 Output pin 11. With the inputs applied respectively as specified in Table III, the output at pin 11 shall be either an output logic “1” or an output logic “0” as specified in Table III.

Table III

Input Pins		Output Pin
8	9	11
0	0	1
0	1	1
1	0	0
1	1	1

3.4.2.4 Output pin 5. With the inputs applied respectively as specified in Table IV, the output at pin 5 shall ’be either an output logic “1” or an output logic “O” as specified in Table IV

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:04Z
Check the source to verify that this is the current version before use.

Table IV

Input Pins		Output Pin
6	7	5
0	0	1
0	1	1
1	0	0
1	1	1

3.4.2.5 Output pin 2. With the inputs applied respectively as specified in Table V, the output at pin 2 shall be as specified in Table V.

Table V

Input Pins		Output Pin
8	9	2
0	0	+18.4 ± 0.5 VDC
0	1	+18.4 ± 0.5 VDC
1	0	0.02 VDC to +1.2 VDC
1	1	+18.4 ± 0.5 VDC

3.4.3 Sequence and timing. With the inputs applied in the sequence specified in Table VI, there shall be the outputs specified in Table VI at pin 3 and at pin 12.

Table VI

Step	Input Pins				Output Pins	
	8	9	15	16	12	3
1	0	1	**	1	0	high
2	0	1	1	0	1*	high
3	1	0	1	0	1	high
4	1	1	1	0	1	low
5	1	1	1	1	0	low
6	1	1	**	1	0	high

* = Logic "1" delayed 200 ± 25 milliseconds from the time pin 16 becomes a-logic "0".

** = Apply a "O" for 1 to 5 seconds and then apply a "1"

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:04Z
Check the source to verify that this is the current version before use.

3.4.4 Capacitor C2. With the signal shown in Figure 2 applied through a 100 ohm ± 10 percent resistor, between pin 13 (+) and pin 1, the voltage out between pin 13 (+) and pin 1 shall be as shown in Figure 3.

3.4.5 Capacitor Cl. With the signal shown in Figure 2 applied through a 100 ohm ± 10 percent resistor, between pin 25 (+) and pin 1, the voltage out between pin 25 (+) and pin 1 shall be as shown in Figure 3.

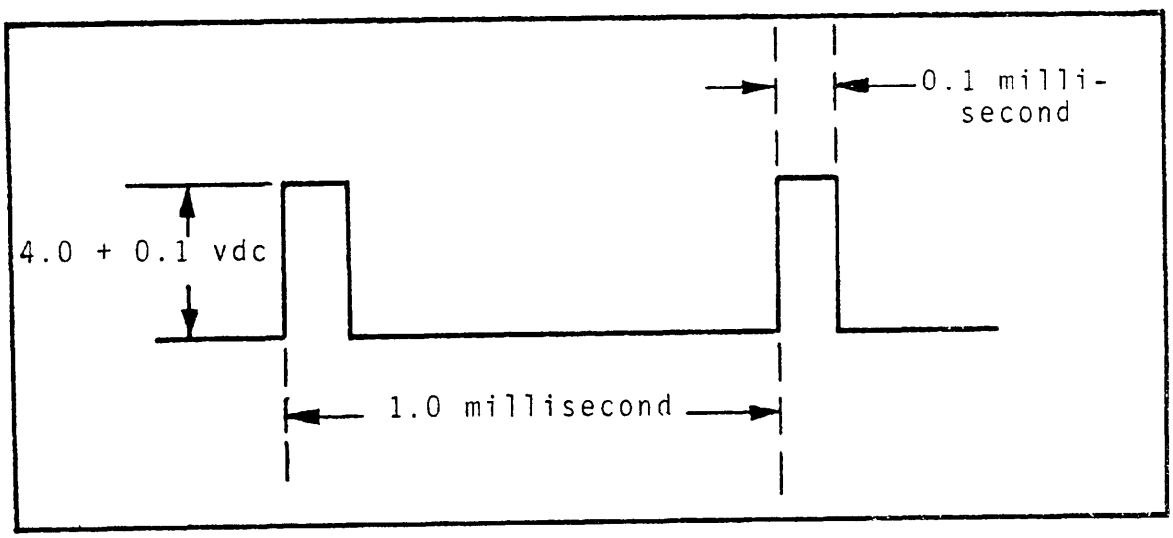


Figure 2

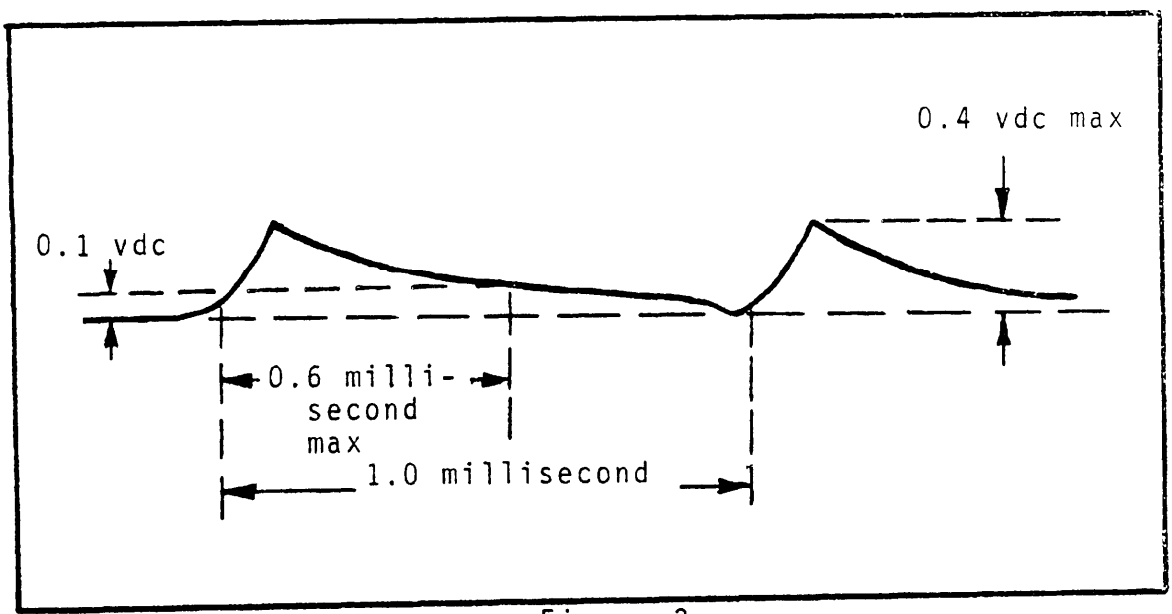


Figure 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:04Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. The quality assurance provisions of MIL-C-60898 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. As specified in MIL-C-60898.

4.3 Control test sample. As specified in MIL-C-60898.

4.4 Functional tests, The following characteristics shall be 100 percent tested in accordance with MIL-C-60898.

	Characteristic	Requirement	Procedure
100	Conditions of Operation	3.4.1	4.5.2.1
101	Logic outputs	3.4.2	4.5.2.2
102	Output pin 27	3.4.2.1	4.5.2 .2.1
103	Output pin 19	3.4.2.2	4.5,2.2.2
104	Output pin 11	3.4.2.3	4.5.2.2.3
105	Output pin 5	3.4.2.4	4.5.2.2.4
106	Output pin 2	3.4.2.5	4.5.2.2.5
107	Sequence and timing	3.4.3	4.5.2.3
108	Capacitor, C2	3.4.4	4.5.2.4
109	Capacitor, Cl	3.4.5	4.5.2.5

4.4.1 Inspection equipment. As specified in MIL-C-60898 and 4.4.1.1.

4.4.1.1 Test equipment. In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

	Item	Description
1.	POWER SUPPLY	22.0 ± 0.1 vdc, 500 ma or greater. Ripple less than 100 mv rms.
2.	POWER SUPPLY	4.5 ± 0.1 vdc, 500 ma or greater. Ripple less than 100 mv rms.
3.	POWER SUPPLY	18.9 ± 0.1 vdc, 500 ma or greater. Ripple less than 100 mv rms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:04Z
Check the source to verify that this is the current version before use.

4.	POWER SUPPLY	0.50 ± 0.01 vdc, 50 ma or greater. Ripple less than 5 mv rms.
5.	POWER SUPPLY	2.80 ± 0,01 vdc, 50 ma or greater. Ripple less than 5 mv rms.
6.	DIGITAL VOLTMETER	With accuracy and resolution of ± 0.2 percent or better from 0.0090 to 99.0 vdc. Input impedance 10 megohms or greater on all voltage scales.
7.	OSCILLOSCOPE	Voltage and time accuracy and resolution of ±5 percent or better. Rise time 100 nanoseconds or less; sweep periods from 1 microsecond to 1 second per centimeter. Amplitude scales from 0.01 to 10 volts/centimeter, automatic triggering.
8.	ASSORTED COMPONENTS	Three RESISTORS 350 ohm ± 5 percent,5 watt. Two CAPACITORS 0.0047 uf ± 20 percent 50 vdc One RESISTOR 100 ohms ± 10 percent, ½ watt.
9.	PULSE GENERATOR	Frequency - 1 kilohertz minimum. Duty factor - 10 percent minimum. Amplitude - minimum of 4 volts into a 200 ohm load. Pulse rise and fall times - less than 0.2 microseconds.

4.5 Test methods and procedures

4.5.1 Environmental. As specified in MIL-C-60898 except the requirements of 3.3.1 herein shall be verified during temperature extremes.

4.5.2 Electrical tests

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:04Z
Check the source to verify that this is the current version before use.

4.5.2.1 Conditions of operation, Unless otherwise specified, all subsequent tests are to be performed with the conditions of operation specified in 3.4.1. Referring to Figure 1, connect R1, R2, R3, Cl and C2, using the ASSORTED COMPONENTS, item 8 of 4.4.1.1. Connect the POWER SUPPLIES, items 1, 2, and 3 of 4.4.1.1, to supply +22.0 ± 0.1 vdc, +4.5 ± 0.1 vdc, and +18.9 ± 0.1 vdc, respectively, with their negative terminals tied to GROUND as shown in Figure 1. When required during subsequent tests, use the positive output of the POWER SUPPLY, item 4 of 4.4.1.1, as logic "0" and the positive output of the POWER SUPPLY, item 5 of 4.4.1.1, as logic "1" Connect the negative terminals of these two supplies to GROUND of Figure 1.

4.5.2.2 Logic outputs. In the following tests, use the DIGITAL VOLTMETER, item 6 of 4.4,1.1, to verify that the logic outputs are within the tolerances specified in 3.4.2.

4.5.2 .2.1 Output pin 27. Referring to Table I, apply logic inputs as shown for pins 22 and 23. Verify that the output voltage on pin 27 conforms to the requirements of Table I and 3.4.2 for each combination of inputs on pins 22 and 23.

4.5. 2.2.2 Output pin 19. Referring to Table II, apply logic inputs as shown for pins 20 and 21. Verify that the output voltage on pin 19 conforms to the requirements of Table II and 3.4.2 for each combination of inputs on pins 20 and 21.

4.5.2. 2.3 Output pin 11. Referring to Table III, apply logic inputs as shown for Pins 8 and 9. Verify that the output voltage on pin 11 conforms to the requirements of Table III and 3.4.2 for each combination of inputs on pins 8 and 9.

4. 5.2.2.4 Output pin 5. Referring to Table IV, apply logic inputs as shown for pins 6 and 7. Verify that the output voltage on pin 5 conforms to the requirements of Table IV and 3.4.2 for each combination of inputs on pins 6 and 7.

4. 5.2.2.5 Output pin 2. Referring to Table V, apply logic inputs as shown for pins 8 and 9. Verify that the output voltage on pin 2 conforms to the requirements of Table V and 3.4.2 for each combination of inputs on pins 8 and 9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:04Z
Check the source to verify that this is the current version before use.

4.5.2.3 Sequence and timing. Referring to Table VI, apply logic inputs as shown for pins 8, 9, 15 and 16 for each step of Table VI. Connect the OSCILLOSCOPE, item 7 of 4.4.1.1, between pin 12 (+) and pin 1. Connect the OSCILLOSCOPE trigger input between 16 and pin 1. Adjust the OSCILLOSCOPE to trigger when pin 16 goes from a logic “l” to a logic “O”. Verify that the output voltage on pins 12 and 3 conform to the requirements of Table VI and 3.4.2 for each specified combination of inputs on pins 8, 9, 15 and 16. Using the OSCILLOSCOPE, confirm Step 2 of Table VI by verifying that pin 12 goes from logic “O” to logic “l” within 200 ± 25 milliseconds after pin 16 is changed to a logic “O” with all other input pins at their required logic level.

4.5.2.4 Capacitor C2. Connect the PULSE GENERATOR, item 9 of 4.4.1.1, through the 100 OHM RESISTOR, item 8 of 4.4.1.1, and connect channel A vertical input of OSCILLOSCOPE, item 7 of 4.4.1.1, between pin 13 (+) and pin 1 per Figure 4. Adjust the PULSE GENERATOR to produce the pulse of Figure 2. Connect the OSCILLOSCOPE channel B vertical input, per Figure 4, and verify the waveform of Figure 3.

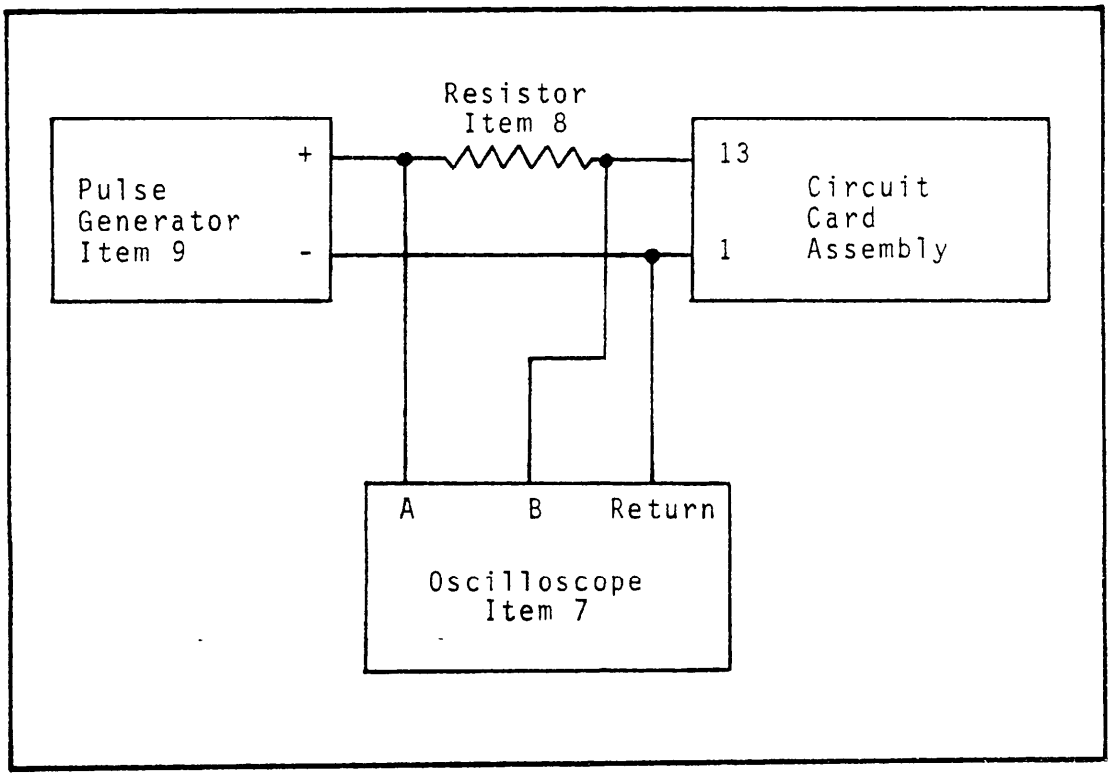


Figure 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:04Z
Check the source to verify that this is the current version before use.

4.5.2.5 Capacitor Cl. Connect the PULSE GENERATOR, item 9 of 4.4.1.1, through the 100 OHM RESISTOR, item 8 of 4.4.1.1, and connect channel A vertical input of OSCILLOSCOPE, item 7 of 4.4.1.1, between pin 25 (+) and pin 1 per Figure 5. Adjust the PULSE GENERATOR to produce the pulse of Figure 2. Connect the OSCILLOSCOPE channel B vertical input, per Figure 5, and verify the waveform of Figure 3,

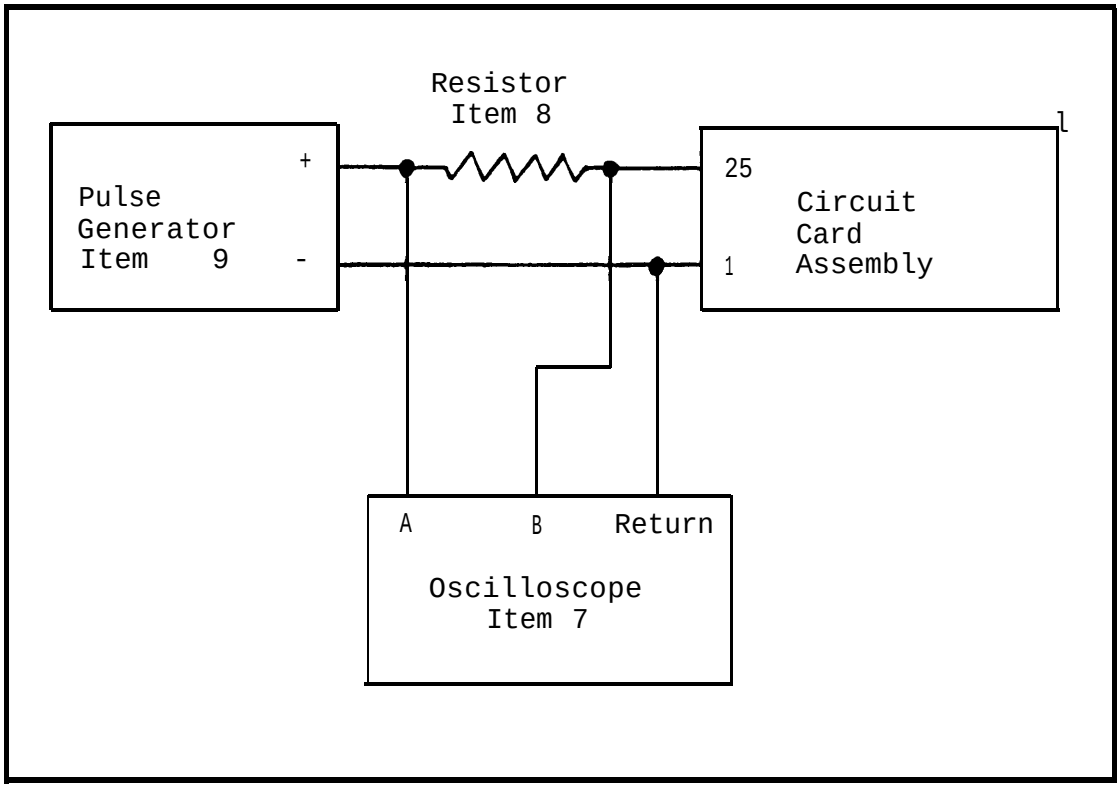


Figure 5

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:04Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with Packaging Data Sheet 10554179.

6. NOTES

6.1 The notes specified in MIL-C-60898 are applicable to this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No.4931-A042

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:04Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:04Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army

Frankford Arsenal
Philadelphia, Pa. 19137

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
Department of the Army

Commanding Officer
Frankford Arsenal
ATTN: SMUFA-N1100
Philadelphia, Pa. 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:04Z
Check the source to verify that this is the current version before use.